

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Concept Mapping

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	K.Madhumitha	Reg. No.:	192513069
Section / Batch:	BE EEE	Date:	10/07/2026

Code of Conduct

I, K.Madhumitha (192513069) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Madhumitha

Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	2	20
Search Data Structure	Linear Search and Binary Search	3	20
Recursion, Performance analysis	Recursion	4	20
Tree Data Structures	Classifications of Tree data structure	5	20
GRAND TOTAL:			100 Marks

Annexure A – Concept Mapping

1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates. (20 Marks)
2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing. (20 Marks)
3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)
4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them. (20 Marks)
5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data. (20 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

STUDENT ATTENDANCE SYSTEM

Records daily presence data of students.

ARRAY

- Stores attendance records in contiguous memory locations.

FIXED SIZE

- Number of records is decided before execution.
- Cannot easily increase or decrease the size.

SEQUENTIAL ACCESS

- Records are accessed one after another.
- Searching may require checking each entry.

LINKED LIST

- Records are stored in separate nodes.
- Each node points to the next node.

DYNAMIC ALLOCATION

- Memory is allocated during program execution.
- New records can be added or removed easily.

FLEXIBILITY

- Efficient for frequent attendance updates.
- Handles insertions, deletions and changing class size easily.

JUSTIFICATION

For dynamic data updates (frequent additions/deletions of students), Linked List is more suitable than Array because it provides dynamic memory and flexibility.

CONCLUSION

A Linked List is the better choice for a student attendance system with changing data.

WEATHER MONITORING SYSTEM

Collects temperature, humidity, wind and other weather data continuously.

01

PRIMITIVE DATA TYPES

- Stores simple sensor readings.
- Examples: int, float, char, boolean.

- Used to store individual values like temperature, humidity etc.
- Occupies less memory and processed quickly.

02

NON-PRIMITIVE DATA STRUCTURES

- Used to store and organize multiple weather records.
- Examples: Array, Linked List, Tree.

- Can store large amounts of data.
- Helps in efficient data management and easy processing.

03

LINEAR STRUCTURES

- Stores data in a sequential order.
- Examples: Array, Linked List, Queue.

- Useful for storing continuous sensor readings.
- Easy to implement and access data in order.

04

NON-LINEAR STRUCTURES

- Stores data in hierarchical or network form.
- Examples: Tree, Graph.

- Represents relationships between multiple weather stations.
- Efficient for complex data analysis and searching.

05

APPLICATIONS

- Used in real-world weather services.
- Examples: Forecasting, Alerts, Agriculture, Aviation, Disaster Management.

- Helps in weather forecasting and prediction.
- Supports decision making in various fields.

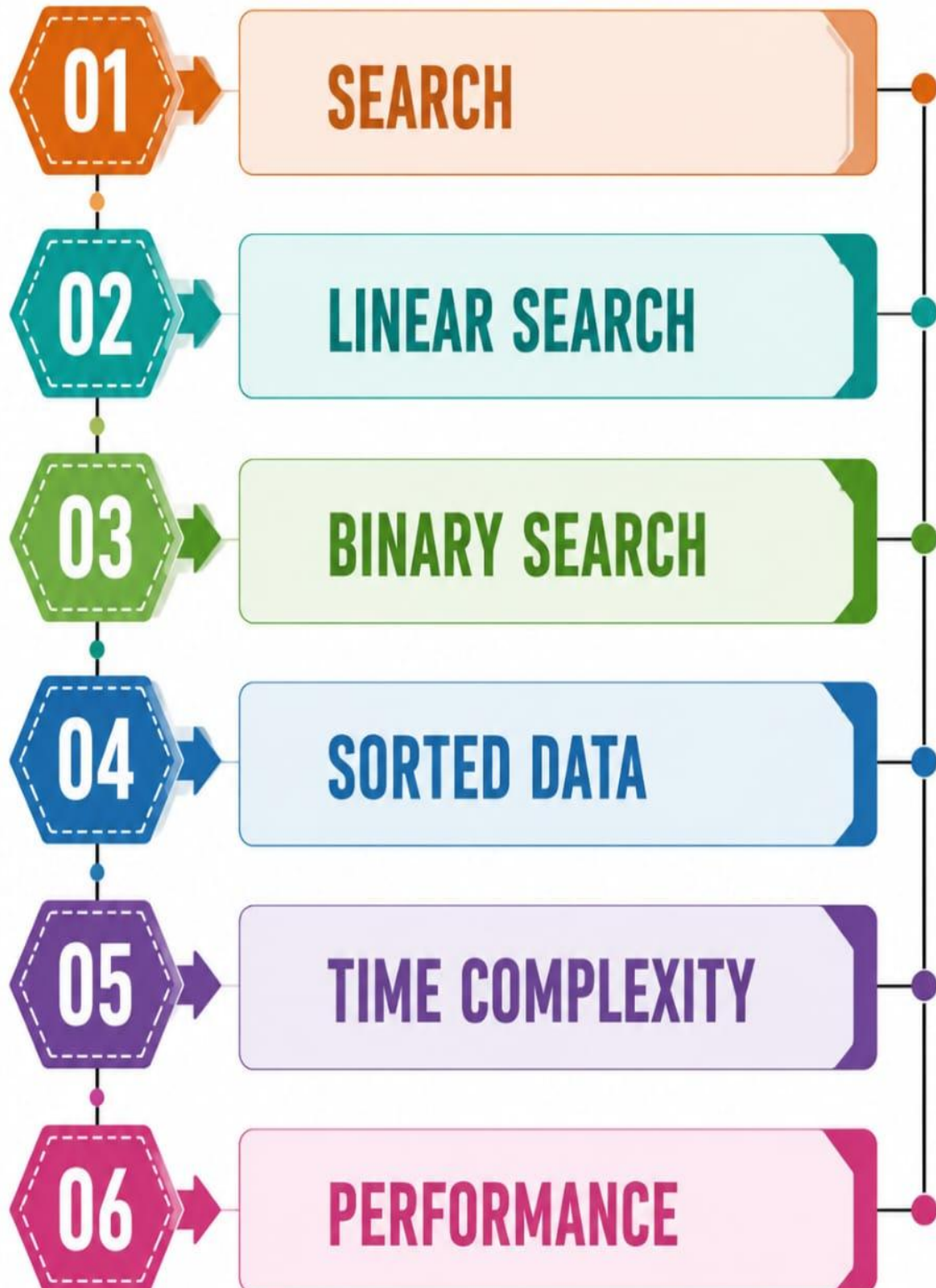
JUSTIFICATION

Both linear and non-linear data structures are necessary. Linear structures store continuous weather data efficiently, while non-linear structures help in analyzing and managing complex relationships between data.

CONCLUSION

Using primitive and non-primitive data structures together allows a weather monitoring system to store large data, process it quickly, and provide accurate forecasts and alerts.

DIGITAL SEARCH SYSTEM





BEST APPROACH

Use Tree structure with appropriate Traversal (DFS/BFS) and Search techniques to efficiently retrieve and manage hierarchical employee records.

